

# RAG over EU Law

Ask anything about EU law. The answer is grounded in the acts in force, every claim is cited, and a citation opens the official PDF with the passage highlighted.

40,183

EU legal documents

all EU law in force, from EUR-Lex

841,805

Chunks

split by article and recital

RAGAS · 100 GOLDEN QUESTIONS

Faithfulness 0.98

Answer relevancy 0.85

Context recall 0.75

Context precision 0.69

FEATURES

pgvector HNSW

LangChain + Gemini

Voyage rerank-3

Structure-aware chunking

Redis exact + semantic cache

SSE streaming

PDF citations with highlights

RAGAS evaluation

LangSmith tracing

Live analytics

Rate limiting

Docker + Caddy

GitHub Actions CI/CD

DEVELOPMENT COST

€3.46

|                                                        |       |
|--------------------------------------------------------|-------|
| RAGAS evaluation after each change                     | €3.46 |
| Oracle Cloud free tier · Ampere, 2 cores, 11 GB        | €0    |
| Voyage rerank-3 · 200M free tokens, used in evaluation | €0    |
| Embedding all chunks · local model                     | €0    |

TRY IT LIVE

rag.elsayed2002.tech



# Answers backed by the actual law.

The screenshot shows the RAG at Scale application interface. On the left, a sidebar contains a 'New chat' button and a search bar. The main chat area displays a query: 'What compensation do air passengers get when their flight is cancelled?'. The application provides a detailed answer based on Article 7 of Regulation (EC) No 261/2004, listing compensation amounts: EUR 250 for flights of 1,500 km or less, EUR 400 for intra-Community flights between 1,500 and 3,500 km, and EUR 600 for all other flights. The answer also mentions that compensation can be reduced by 50% if the passenger is re-routed. At the bottom, sources are listed as 'Regulation (EC) No 261/2004', page 4 and page 5. On the right, a preview of the source document is shown, highlighting the relevant legal text.

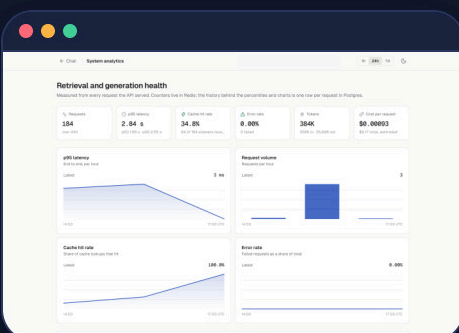
## THE ANSWER SAYS

"EUR 250 for all flights of 1,500 kilometres or less <sup>1</sup>"



## THE LAW SAYS

The screenshot shows the official text of Article 7 of Regulation (EC) No 261/2004, titled 'Right to compensation'. It specifies the compensation amounts for flight cancellations: EUR 250 for flights of 1,500 km or less, EUR 400 for intra-Community flights between 1,500 and 3,500 km, and EUR 600 for all other flights. It also states that compensation can be reduced by 50% if the passenger is re-routed within certain timeframes.



## Live analytics

Every request is measured and stored. The dashboard shows:

p50 · p90 · p95 latency

Latency distribution

Time per step

Cache hit rate

Error rate

Tokens in / out

Cost per request

Requests per hour

Slowest queries

Repeated questions

1 h · 24 h · 7 d

# Chunk by legal structure, not by size.

1

## From PDFs to legal structure

Hit@6: right passage among the top 6 · same model, only the chunking differs

Version 1 · 43,106 PDFs

**Recursive splitting, ~1,600 characters**

200 overlap · 1,062,613 chunks

Hit@6 **32%**

on 351,858 chunks



Version 2 · 40,183 HTML documents

**One chunk per article and recital**

841,805 chunks

Hit@6 **46%**

on 330,850 chunks

2

## Every chunk knows where it is in the law

the Article 7 chunk behind the answer on page 2

Regulation (EC) No 261/2004 ... | Article 7  
Right to compensation

1. Where reference is made to this Article, passengers shall receive compensation amounting to:

(a) EUR 250 for all flights of 1500 kilometres or less;

(b) EUR 400 for all intra-Community flights of more than 1500 kilometres, and for all other flights between 1500 and 3500 kilometres;

(c) EUR 600 for all flights not falling under (a) or (b). ...

**Articles:** one chunk each. Long ones are split at their numbered paragraphs (max 480 tokens).

**Recitals:** a few per chunk (max 250 tokens), labelled like "Recitals (14)-(15)".

**Tables of numbers** in annexes are skipped: they match no question.

# 841,805 vectors: embedded for free, searched in milliseconds.

1

## Embedding

turning 841,805 chunks into vectors

Gemini · free tier

2.3 years

1,000 per day

Gemini · paid

~\$55

every re-embedding

bge-base · chosen

\$0 · 1 hr

free Kaggle GPU

Gemini is more accurate, but every new chunking means paying again. So I kept the free model and improved the other steps.

2

## Vector database: building HNSW

PostgreSQL + pgvector · so a question isn't compared with every vector

Without an index

5.6 s

compares the question with every vector



With the HNSW index

12 ms

walks a graph to the nearest ones

460×

faster

99% the same results

### Tuning the index



Built in 14 minutes

1.6 GB, kept in memory

16-bit vectors: half the size

# Finding the right passage.

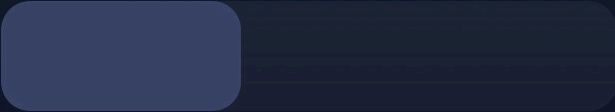
1

## Reranking

Hit@6 over all 841,805 chunks: right passage among the 6 results the model reads

### Vector search alone

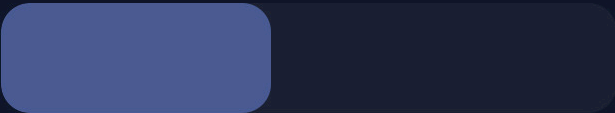
the 6 chunks closest in meaning



39%

### + at most 1 recital

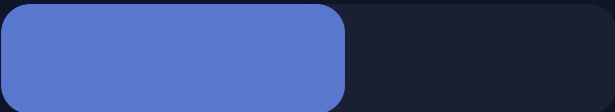
recitals explain why a law exists; they crowded out the rules



44%

### + local reranker, top 20

a small model re-reads the best 20 with the question



56%

### + Voyage rerank-3, top 100

a stronger model re-reads the best 100



80%

Vector search finds the right passage somewhere in its top 100 for **94%** of questions. The reranker moves it into the top 6 for **80%**.

2

## How the system was measured

every change tested before it shipped

**110 golden questions**  
with verified answers, 10  
of them out of scope

**Hit@6** after every  
change: free, no LLM  
needed

**RAGAS** with an LLM  
judge for the final  
answers

# Running it for real.

## 1 Caching: fast answers, lower cost

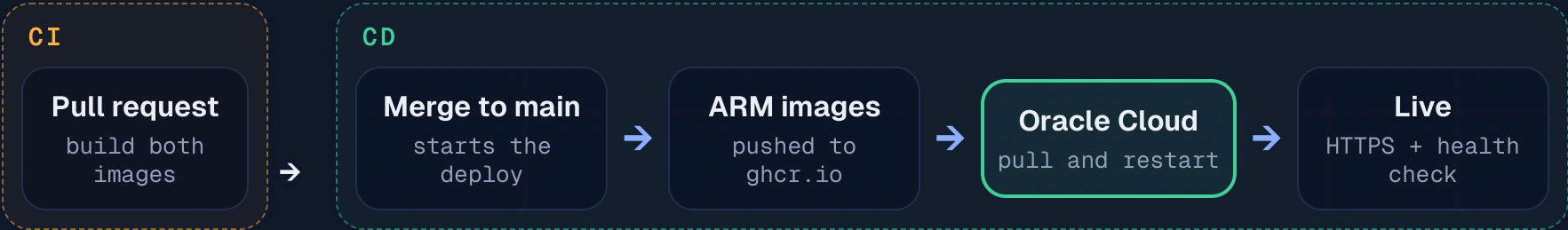
measured live, from outside, over 170 requests

| First words         | Full answer | Same question | Similar question |
|---------------------|-------------|---------------|------------------|
| 1.4 s               | 2.1 s       | 76 ms         | 201 ms           |
| streamed as written | median      | exact · \$0   | semantic · \$0   |

A cached answer skips search, reranker and Gemini: it costs nothing. Similar questions are reused only at  $\geq 0.95$  similarity ("wage in 2026" vs "2025" is 0.93).

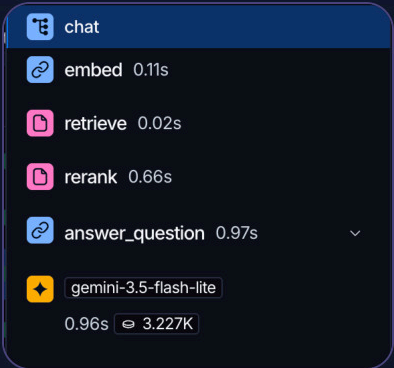
## 2 CI/CD pipeline

GitHub Actions · every merge to main goes live in ~5 minutes



## 3 Observability

every answer traced in LangSmith



Every step of every answer is logged. We can see each step's **input** and **output**.

Every request logged   Time per step

Try it live  
[rag.elsayed2002.tech](https://rag.elsayed2002.tech)

Source code  
[github.com/mostafaelsayed2002/rag-at-scale](https://github.com/mostafaelsayed2002/rag-at-scale)